CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

ITUC201 — Fundamentals of Computer Networks | Student: Bhavy Gol (ID: 25CS016)

UNIVERSITY END-SEM EXAMINATION QUESTION & WORKED SOLUTION

MAIN QUESTION (Q1 — 7 MARKS)

A sender application at the Data Link Layer needs to transmit the binary payload **011111101111101** over an HDLC synchronous link.

1. Show step-by-step sender bit-stuffing.
2. Construct the complete HDLC-style framed bit stream.
3. Demonstrate receiver destuffing and payload restoration.
4. Calculate bit stuffing overhead (bits and percentage).
5. Justify why bit stuffing is mandatory in HDLC.

Part 1: Step-by-Step Sender Bit-Stuffing Audit Trace

Step #	Input Bit	Consecutive 1s	Action Taken	Stuffed Output Stream
1	0	0	Keep	0
2	1	1	Keep	01
3	1	2	Keep	011
4	1	3	Keep	0111
5	1	4	Keep	01111
6	1	5	Stuff '0'	0111110
7	1	1	Keep	01111101
8	0	0	Keep	011111010
9	1	1	Keep	0111110101
10	1	2	Keep	01111101011
11	1	3	Keep	011111010111
12	1	4	Keep	0111110101111
13	1	5	Stuff '0'	011111010111110
14	0	0	Keep	0111110101111100
15	1	1	Keep	01111101011111001

Part 2: HDLC-Style Framed Stream

Framed Stream = FLAG | STUFFED PAYLOAD | FLAG

01111110 01111101011111001 01111110 (Total Length = 33 bits)

Part 3: Receiver Destuffing Verification

Receiver strips flags, isolates stuffed payload 01111101011111001. Discards inserted zeros after 5 consecutive 1s.
Restored Payload: 011111101111101 (STATUS: PASS 100% Match)

Part 4: Stuffing Overhead Calculation

Inserted Zeros = 2 bits | Original Length = 15 bits
Stuffing Overhead (%) = (2 / 15) * 100% = 13.33%

Part 5: Technical Justification

Bit stuffing guarantees data transparency by ensuring payload bits can never contain six consecutive 1-bits, preventing premature frame delimiter detection.

ADDITIONAL PRACTICE QUESTIONS (WITH ANSWERS)

Q2: Stuff payload '1111111111' (ten ones).
Ans: Stuffed = 111110111110 (12 bits) | Overhead = 20.00%

Q3: Destuff receiver stream '1111100111110'.
Ans: Restored payload = 11111011111 (11 bits)